Streaming Experience Observability

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One system that goes from impacted viewers — how many, where, what broke — to why, by lining player QoE and backend services up on the same minute, at a fraction of the cost of today's disconnected tools.

The problem

Large publishers find out that viewers are suffering from one tool and why from another, and a person has to join the two during the incident.

- **Three or four disconnected products.** A QoE vendor for viewers, an APM vendor for services, CDN logs in a SIEM, a warehouse for analysis. Each has its own copy of the data, its own clock and its own name for the same CDN edge.
- **Priced on the wrong quantity.** QoE tools scale with player events. APM scales with hosts and ingested spans. Neither cost tracks what matters, which is viewing minutes and the critical delivery path.
- **Slow answer to "why".** Alerts say a metric moved. Finding the responsible cohort and service is manual: screenshots, war rooms, hours.
- **History is expensive or gone.** Minute-level detail is kept for days or weeks. Ad-hoc questions run on the same costly clusters that do monitoring, or are not asked at all.

The solution

Measure exactly where the truth is, summarize to one record per minute, and keep raw data only as evidence. Players and services follow the same rule.

1. **Player SDK.** Measures exact timings on the device and sends one small record per viewer per minute (~5 KB per viewer-hour, versus ~95 KB for event streams).
2. **Services lane.** An OpenTelemetry agent on each critical-path node (origin shield, packager, license, ads) turns all of its spans into a per-minute summary on the same UTC clock, the same latency buckets and the same CDN-edge and region names.
3. **Detection and attribution.** Bounded cubes are checked against seasonal, mix-aware baselines. Cohorts are ranked by how much excess impact they explain, not by worst rate.
4. **Aligned view.** Every alert shows which service path moved with the impacted cohort, in the same minute, and which moved first. Player timelines and traces are pulled only for that cohort.

5. **Cheap analytics path.** Each hour, history lands as Parquet on object storage. Any slice loads into a local DuckDB for analysts or an analytics agent, so ClickHouse stays sized for monitoring and troubleshooting.

Question	Answered by
How many viewers, how badly?	Impacted viewing seconds, content and ads separately
Where?	Impact-ranked cohorts: CDN, ISP, region, device, app
What?	Rebuffer, stall-out, startup, errors, DRM, ads, quality
Why?	The service path that moved with that cohort in the same minute
Show me	Player timelines and traces for that cohort only

The value

Faster answers to "why", on one source of truth, with infrastructure that grows with viewing minutes instead of events. Planning figures at 1M concurrent viewers:

Lever	Today	With this system
Player telemetry	~95 KB per viewer-hour; ~14 TB/day	~5 KB per viewer-hour; ~115 GB/day (~19x less)
Critical-path service telemetry	~14.6 MB/s of spans shipped and indexed	~23 KB/s of minute summaries; spans leave the node only as evidence
Viewer-to-service correlation	Manual, across tools, during the incident	Attached to every alert, inside the 5-minute detection budget
History for ad-hoc analysis	Days to weeks at minute detail, on costly clusters	13 months of minute detail, about 5 TB on object storage (about $100-120/month)

- **Shorter incidents.** On-call goes straight to the cohort and service path that explain the impact, instead of the worst-looking but small cohort.
- **Lower bill, where it is retired.** The QoE licence and the critical-path share of APM span ingest are the line items that shrink.

- **Questions nobody can afford to ask today.** For example, a year of rebuffering by title, or replaying last quarter with a new threshold. Analysts and agents work on a laptop, not the production cluster.

What must be proven

All figures above are planning arithmetic until a shadow phase measures them on production traffic.

- **SDK parity and coverage.** Minute records reproduce trusted metrics on web and one native platform, with acceptable loss on CTV.
- **Cube width.** Real context cardinality keeps the cubes small enough to scan every 30 seconds.
- **Edge mapping.** The CDN sends its edge ID on origin fetches, and one shared edge dimension maps both sides.
- **Attribution quality.** Injected and historical incidents are found with few false alarms.
- **Total cost.** Savings depend on retiring licences and span ingest. Building still needs a dedicated team across SDKs, backend and on-call.